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ReST-Net: Diverse Activation Modules and Parallel Sub-nets based CNN for Spatial Image Steganalysis

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Abstract—Recent steganalytic schemes reveal embedding traces in a promising way by using Convolutional Neural Networks (CNNs). However, further improvements, such as exploring complementary data processing operations and using wider structures, were not extensively studied so far. In this paper, we design a new CNN in these aspects in order to better capture embedding artifacts. Specifically, on the one hand, we propose to process information diversely with a module called diverse activation module. On the other hand, we build a wide structure with parallel sub-nets using several filter groups for pre-processing. To accelerate the training process, we pre-train the sub-nets independently and feed their output vectors together to a classification module. Extensive experiments show that the proposed method is effective in detecting content-adaptive steganographic schemes.

Index Terms—Steganalysis, steganography, convolutional neural networks, diverse activation, wide structure

I. INTRODUCTION

Covert communication with steganography [1] has attracted considerable attention in recent years. Modern image steganographic schemes, such as [2]–[6], are considered safer as they hide information in an adaptive way so that it is difficult to detect their embedding traces. Steganography can be used for criminal purpose [7], [8]. Consequently, detecting steganography, also called steganalysis [9], is an important task.

Most of steganalytic approaches were based on hand-crafted features allied to machine learning classifiers [10]–[14]. This reality started to change since the rise of Convolutional Neural Networks (CNNs) [15]–[18]. Such data-driven methods have achieved satisfactory performance in image classification. Several CNN solutions tried to address the spatial image steganalysis problem [19]–[24]. However, most of these schemes use only one learning pipeline. Further improvements, such as using wider structures, were not extensively explored so far.

In this letter, we propose a more effective data-driven CNN structure for steganalysis. On the one hand, inspired by the *Inception module* [17] which increases the width of CNN architecture with convolutional kernels of different sizes, we propose a CNN architecture composed of *diverse activation modules*, which activate the convolution outputs differently and then concatenate their outputs for the subsequent layers. This way, more clues of data embedding are passed through the network, making the detection of embedding traces more accurate. Since ReLU, Sigmoid, and TanH activation functions

are used, we name our network as ReST-Net. On the other hand, inspired by the diverse sub-models used in Spatial Rich Models (SRMs) [11], we construct a structure with multiple sub-nets, whose inputs are pre-processed with several groups of high-pass filters. Consequently, the embedding traces are highlighted in a complementary way. In the experiments using the BOSSBase image set [25], we show that the proposed approach outperforms the established methods.

The remaining of this letter is organized as follows. In Section II, we review some data-driven approaches for image steganalysis. We present our CNN architecture with multiple activation units and parallel sub-nets in Section III. In Section IV, we introduce the experimental settings and report the results. This letter is concluded in Section V.

II. RELATED WORK

The first work to consider deep learning structures for image steganalysis was done by Tan and Li with convolutional auto-encoders [19]. Qian *et al.* [20] proposed a CNN structure comprised of a pre-processing layer equipped with a high-pass filter. The performances of these schemes are comparable to or better than the feature engineering based method called SPAM (Subtractive Pixel Adjacency Matrix) [10], but are still inferior to the well established SRM scheme [11]. A breakthrough was achieved by Xu *et al.* [21]. Their method (called Xu-CNN in this paper) is comparable with the hand-crafted SRM feature based method for the first time. The authors considered several advanced CNN techniques used in image classification tasks, such as batch normalization (BN) [26], 1×1 convolution, and global average pooling [27]. They also applied steganalytic domain knowledge based operations, such as pre-processing with a high-pass filter and using an absolute (ABS) activation layer. By using an ensemble of a modified version of Xu-CNN, a more stable performance can be achieved [22]. By mimicking the process of another effective hand-crafted feature based steganalytic method, called PSRM (Projection Spatial Rich Model) [12], Sedighi and Fridrich [23] proposed a CNN structure with histogram layers, formed by a set of mean-shifted Gaussian activation functions. Ye *et al.* [24] proposed a CNN structure with a group of high-pass filters for pre-processing and adopted a new activation function, called truncated linear unit (TLU), to better capture the embedding signals. It has been shown that a wider architecture may improve the performance of CNNs. However, it has not been extensively explored in steganalysis so far. This motivates us to design a wider CNN, with more parallel activation units and processing pipelines. We will discuss our proposed method in the next section.

III. THE PROPOSED METHOD

A. Architecture Overview

The proposed CNN structure, as shown in Fig. 1, is composed of three parallel convolutional sub-nets and a fully-

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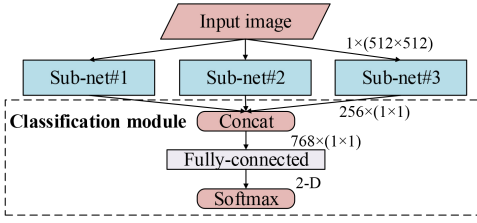


Fig. 1. The proposed CNN architecture with three parallel sub-nets and a classification module.

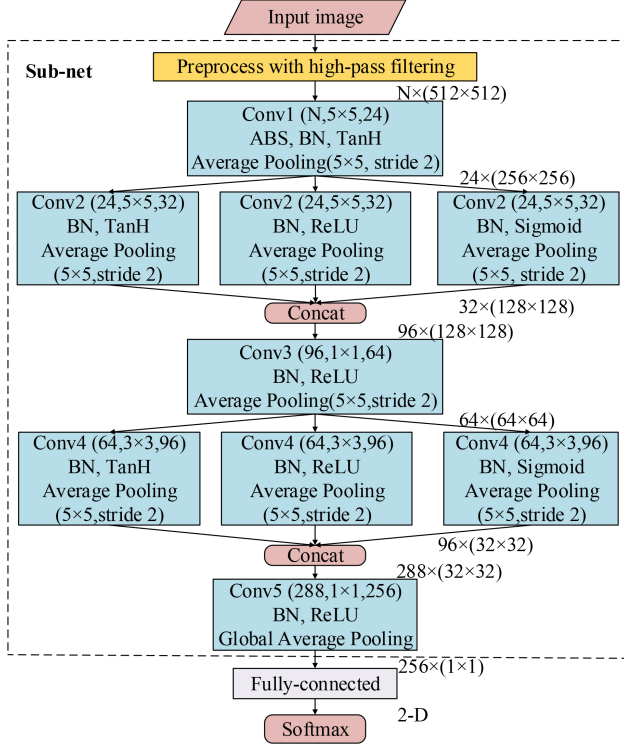


Fig. 2. The proposed pre-trained sub-net with DAM. A convolutional group is represented by a blue box. N is the number of filter residuals. $\text{Conv}(x_1, a \times a, x_2)$ denotes the convolution layer with the kernel size $a \times a$ for x_1 input feature maps and x_2 output feature maps. Batch normalization is abbreviated as BN. Absolute activation is abbreviated as ABS.

connected classification module. Each sub-net accepts an input image of size 512×512 and outputs a 256-D feature vector. These sub-nets act as data-driven feature extractors, and are built based on Xu-CNN [21] but equipped with *diverse activation modules* (DAMs) in some convolutional groups, which are further described in Section III-B. The only difference among sub-nets is the way they pre-process the input, which are explained in Section III-C.

The proposed CNN is trained in two phases. In the first phase, each sub-net is pre-trained independently with a fully-connected layer and a Softmax function, as shown in Fig. 2, to classify cover and stego images. Once the pre-training is done, the parameters in the sub-nets are frozen without further training, and the fully-connected layers are discarded. In the second phase, a new fully-connected layer with 768 (256×3) input neurons is fed with the concatenated output feature vectors from the final convolutional groups of all three sub-nets. Such a fully-connected layer with a Softmax function is trained and acts as the final classification module. This training process with two phases, *i.e.*, training the sub-nets for feature extraction, and training the fully-connected layer for classification, can ensure stable and efficient convergence

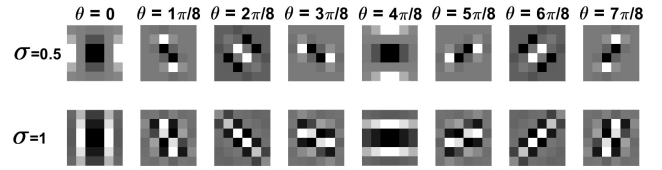


Fig. 3. Illustration of Gabor filters with different orientations and scales.

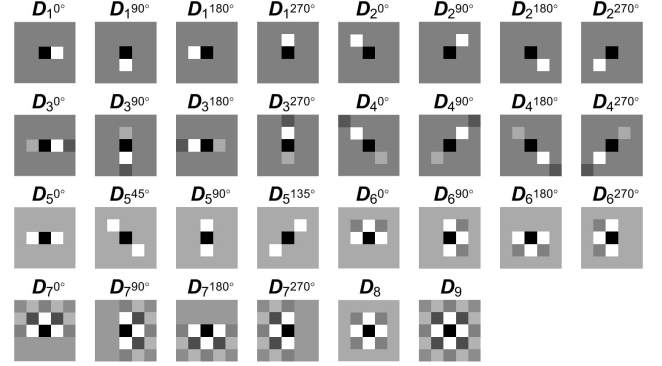


Fig. 4. Illustration of SRM filters.

compared with training the proposed CNN as a whole without pre-training.

B. Diverse Activation Modules

It has been widely accepted that wider networks can carry more significant information through CNN layers. For example, the *Inception module* [17] increases the width by using convolutional kernels of different sizes. Inspired by the idea of increasing the width to boost performance, we adapt Xu-CNN [21] by using what we call the *Diverse Activation Modules* (DAMs). In Xu-CNN, TanH is used for activation in the first two convolutional groups, and ReLU is used in the last three. From the results reported in [21], when the activation functions are replaced, the performance degrades. In order to learn the steganography artifacts differently, we form DAMs by using ReLU, Sigmoid and TanH activation functions simultaneously in the second and the fourth convolutional groups, and concatenating the resulting feature maps for the subsequent convolutional groups. The proposed sub-net architecture can be seen in Fig. 2. According to the initial letters of the activation units, we call the network as ReST-Net. It is expected that each activation unit in DAMs may have different responses to the embedding traces. The diversity may help to boost classification performance, as shown in the experiment in the next section. Note that the DAM is not used in all convolutional groups because the number of the weights in convolutional kernels may increase, making such a wide network less efficient for convergence. The slim structure for the third and the fifth convolutional groups in the proposed network is similar to the “bottleneck block” in ResNet [18].

C. Parallel Sub-nets with Different Pre-processing Operations

The structure of the three sub-nets are identical, except for their pre-processing layers with high-pass filtering operations, as shown in Fig. 2, where N is the number of the filtered residuals. The high-pass filtering technique is pervasively used in steganalysis [10]–[13] to capture embedding artifacts. CNN based methods [20], [21], [24] also benefits from such a technique. One may speculate that involving more high-pass filters could boost the steganalytic performance. However, as

TABLE I
DETECTION ACCURACIES (IN %) FOR S-UNIWARD, HILL, AND CMD-HILL. DIFFERENCES COMPARED WITH XU-CNN AND TLU-CNN ARE SHOWN IN BOLD AND ITALIC FONTS, RESPECTIVELY.

CNN Scheme	S-UNIWARD (bpp)					HILL (bpp)					CMD-HILL (bpp)				
	0.1	0.2	0.3	0.4	0.5	0.1	0.2	0.3	0.4	0.5	0.1	0.2	0.3	0.4	0.5
Xu-CNN [21]	59.43	66.67	73.68	80.12	83.54	58.93	66.75	73.14	78.69	81.82	55.64	61.73	66.75	71.61	74.75
TLU-CNN [24]	59.71	66.49	74.38	77.36	82.36	56.45	65.35	72.02	76.92	78.86	54.95	60.24	68.71	74.38	77.25
Sub-net #1	64.62	69.58	76.13	84.62	85.87	60.54	68.52	74.57	80.44	83.29	56.71	63.43	68.49	74.36	77.25
	+5.19	+2.91	+2.45	+4.50	+2.32	+1.61	+1.77	+1.43	+1.75	+1.47	+1.07	+1.70	+1.74	+2.75	+2.50
Sub-net #2	64.15	68.73	76.44	84.28	86.17	60.88	69.13	75.16	80.25	83.47	56.93	64.15	69.66	74.89	78.37
	+4.72	+2.06	+2.76	+4.16	+2.63	+1.95	+2.38	+2.02	+1.56	+1.65	+1.29	+2.42	+2.91	+3.28	+3.62
Sub-net #3	60.24	66.37	74.75	78.54	83.02	61.23	68.56	74.67	79.59	82.87	56.49	62.94	68.11	73.06	76.52
	+0.81	-0.30	+1.07	-1.58	-0.52	+2.30	+1.81	+1.53	+0.90	+1.05	+0.85	+1.21	+1.36	+1.45	+1.77
ReST-Net	65.67	71.35	78.78	85.44	87.93	62.38	70.64	76.74	81.66	84.54	58.92	65.14	70.28	76.17	79.16
	+6.24	+4.68	+5.10	+5.32	+4.39	+3.45	+3.89	+3.60	+2.97	+2.72	+3.28	+3.41	+3.53	+4.56	+4.41

TABLE II
DETECTION ACCURACIES FOR SUNIWARD, HILL AND CMDHILL(%) BY REPLACING THE DAM IN OUR PROPOSED APPROACH WITH ReLU, SIGMOID, AND TanH, RESPECTIVELY. DIFFERENCES COMPARED WITH THE SCHEME USING DAM ARE SHOWN IN BOLD.

CNN Scheme	S-UNIWARD		HILL		CMD-HILL	
	0.1	0.4	0.1	0.4	0.1	0.4
Sub-net #1 with ReLU	59.81	82.84	58.76	79.25	55.76	73.18
	-4.81	-1.78	-1.78	-1.19	-0.95	-1.18
Sub-net #1 with Sigmoid	63.8	83.65	59.06	77.07	55.12	71.18
	-0.82	-0.97	-1.48	-3.37	-1.59	-3.18
Sub-net #1 with TanH	62.33	83.39	59.31	78.2	56.22	71.15
	-2.29	-1.23	-1.23	-2.24	-0.49	-3.21
Sub-net #2 with ReLU	62.37	83.62	59.43	78.86	55.41	72.74
	-1.78	-0.62	-1.45	-1.39	-1.52	-2.15
Sub-net #2 with Sigmoid	63.44	82.86	59.12	77.84	56.04	71.42
	-0.71	-1.42	-1.76	-2.38	-0.89	-3.47
Sub-net #2 with TanH	63.72	83.81	59.92	78.71	56.43	72.75
	-0.43	-0.49	-0.96	-1.52	-0.50	-2.14
Sub-net #3 with ReLU	57.45	77.29	58.54	78.23	56.17	72.42
	-2.79	-1.25	-2.69	-1.36	-0.32	-0.64
Sub-net #3 with Sigmoid	59.02	76.98	58.32	76.46	55.25	69.77
	-1.22	-1.56	-2.91	-3.13	-1.24	-3.29
Sub-net #3 with TanH	59.02	76.98	58.32	76.46	55.25	69.77
	-1.22	-1.56	-2.91	-3.13	-1.24	-3.29
Ensemble with ReLU	62.93	84.37	60.35	79.86	57.32	73.95
	-2.74	-1.07	-2.03	-1.80	-1.60	-2.22
Ensemble with Sigmoid	65.13	84.59	60.11	78.87	57.16	75.08
	-0.54	-0.85	-2.27	-2.79	-1.76	-1.09
Ensemble with TanH	64.53	84.68	60.6	79.63	58.02	74.83
	-1.14	-0.76	-1.78	-2.03	-0.90	-1.34

shown in our experiments in Section IV-B, simply stacking a large number of high-pass filters cannot perform the best. It is more effective when we use sub-nets equipped with different sets of high-pass filters. Linear filters and non-linear filters from SRM [11], together with Gabor filters [28], are respectively employed in the three sub-nets. The SRM linear filters can capture complicated linear relationships among image pixels, and were effectively used in [24]. The SRM non-linear filters introduce non-linearity and diversity to capture stego abnormality, but were not tried in previous CNN designs. The Gabor filters are complementary to SRM filters, for that they can analyze the image with a specific frequency in a

specific direction. The details of the pre-processing layer for each sub-net are explained as follows.

1) *Sub-net #1*: The input image is pre-processed by filtering with a set of 6×6 Gabor filters [28], and the resulting images are the input of the first convolutional block. The Gabor filter is defined as the product of a Gaussian function and a cosine function as shown in the following equation:

$$g(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(2\pi \frac{x'}{\lambda}\right), \quad (1)$$

where $x' = x \cos \theta + y \sin \theta$, $y' = -x \sin \theta + y \cos \theta$, θ is the orientation of the filter, σ is the scale parameter, $\lambda = \frac{\sigma}{0.56}$ is the wavelength of the cosine function, and $\gamma = 0.5$ is the spatial aspect ratio to specify the Gaussian ellipticity. We use eight orientations (θ from 0 to $7\pi/8$) with two scales ($\sigma = 1$ and $\sigma = 0.5$); consequently the number of filters and residual maps yielded are 16. As done before in [28], all the filters are made zero-mean by subtracting the mean of the filter elements. The filters are illustrated in Fig. 3 for better understanding.

2) *Sub-net #2*: The input image is pre-processed by linear filtering with a set of high-pass filters from SRM [11]. We pad the filters with zeros to obtain a unified size of 5×5 . The filters are renamed and grouped into 9 classes. As shown in Fig. 4, there are four filters with different angles in the first 7 filter classes. We selected some filters for use as follows: $D_1^{0^\circ}$, $D_1^{90^\circ}$, $D_2^{0^\circ}$, $D_2^{90^\circ}$, $D_3^{0^\circ}$, $D_3^{90^\circ}$, $D_4^{0^\circ}$, $D_4^{90^\circ}$, $D_5^{0^\circ}$, $D_5^{90^\circ}$, $D_6^{0^\circ}$, $D_6^{90^\circ}$, $D_7^{0^\circ}$, $D_7^{90^\circ}$, D_8 and D_9 . As a result, 16 linear residual images are obtained and used as the pre-processed inputs for Sub-net #2.

3) *Sub-net #3*: In order to introduce non-linearity, the input image is firstly pre-processed by filtering with some SRM high-pass filters discussed before, and then the resultant residual images are nonlinearly processed with “max” or “min” operation, as done in [11]. Denote R_D the residual image by convolving an image with a filter D . The “max” (or “min”) operator computes the maximum (or minimum) values among the residual images within a filter class. The output residual is denoted by R_D^{max} (or R_D^{min}). For example,

$$R_{D_1}^{max} = \max\{R_{D_1^{0^\circ}}, R_{D_1^{90^\circ}}, R_{D_1^{180^\circ}}, R_{D_1^{270^\circ}}\}, \quad (2)$$

$$R_{D_1}^{min} = \min\{R_{D_1^{0^\circ}}, R_{D_1^{90^\circ}}, R_{D_1^{180^\circ}}, R_{D_1^{270^\circ}}\}. \quad (3)$$

As shown in Fig. 4, the filters in the first 7 classes are processed in this manner. As a result, 14 non-linear residual

TABLE III
CORRELATION COEFFICIENTS BETWEEN THE MEAN GRADIENT VALUES OF DIFFERENT ACTIVATION UNITS.

	Sub-net #2 with DAM			Sub-net #2 with ReLU			Sub-net #2 with Sigmoid			Sub-net #2 with TanH		
Activation #1 Activation #2	ReLU TanH	ReLU Sigmoid	TanH Sigmoid	ReLU1 ReLU2	ReLU1 ReLU3	ReLU2 ReLU3	Sigmoid1 Sigmoid2	Sigmoid1 Sigmoid3	Sigmoid2 Sigmoid3	TanH1 TanH2	TanH1 TanH3	TanH2 TanH3
2nd Conv	0.7890	0.8202	0.9190	0.9759	0.9874	0.9675	0.9613	0.9520	0.9854	0.9584	0.9435	0.9762
4th Conv	0.8575	0.8836	0.9401	0.9886	0.9938	0.9897	0.9740	0.9849	0.9816	0.9730	0.9902	0.9841

TABLE IV
DETECTION ACCURACIES (%) WHEN SIX CASES OF SUB-NETS ARE USED. DIFFERENCES COMPARED WITH ReST-Net ARE SHOWN IN BOLD FONTS.

Case	I	II	III	IV	V	VI
S-UNIWARD 0.1 bpp	61.95 -3.72	60.91 -4.75	64.48 -1.19	63.12 -2.45	64.72 -0.95	67.13 +1.47
S-UNIWARD 0.4 bpp	82.76 -2.68	81.85 -3.59	84.93 -0.51	84.89 -0.55	84.07 -1.37	85.73 +0.29

images are obtained and used as the pre-processed inputs for Sub-net #3.

IV. EXPERIMENTS

A. Setup

We used the BOSSBase v1.01 dataset [25], which contains 10000 uncompressed images of size 512×512 and S-UNIWARD [4], HILL [5], and CMD-HILL [6] for data embedding, with the payload from 0.1 to 0.5 bpp (bit per pixel). The images were randomly split into a training set with 4000 cover and stego image pairs, a validation set with 1000 image pairs, and a testing set containing 5000 image pairs. We used Tensorflow v1.1 for implementation. The network weights were initialized using a normal distribution with zero mean and standard deviation 0.01. The learning algorithm was the minibatch gradient descent with a 0.9 momentum. The initial learning rate was set to 0.001 and the learning decay was set to 90% every 5000 training steps. The batch size was set to 40, with 20 cover images and their corresponding stego counterparts. The training decay was set to 0.05. The epsilon in BN layer was 0.001. We used 1000 and 50 training epochs to train sub-nets and the classification module, respectively. The performance was evaluated by the testing accuracy, where the best validation model obtained during training was selected.

B. Results

1) *Comparison to existing methods:* We conducted experiments to compare the proposed approach with Xu-CNN [21] and the method in [24] without the selection-channel information (which is called TLU-CNN). Table I shows the results. First, it can be observed that in most cases all three individual sub-nets outperformed the considered baseline methods. Considering that 30 high-pass filters are used in TLU-CNN [24] and only 14 to 16 filters are used in the proposed sub-nets, we may conclude that using the DAMs rather than increasing the number of high-pass filters takes effect. Second, ReST-Net had the best improvement in all the experiments. On S-UNIWARD, HILL, and CMD-HILL with different payloads, ReST-Net got an averaged accuracy improvement of 5.14%, 3.32%, and 3.83% over Xu-CNN, and 5.77%, 5.27%, and 2.83% over TLU-CNN, respectively. The fact that the overall ReST-Net performed better than any of the sub-nets justifies the effectiveness of the proposed parallel sub-nets structure.

2) *Further investigation on DAM:* We would like to find out the benefits of using the proposed DAMs. For that, we performed experiments by comparing how the individual sub-nets and the ensemble of these sub-nets were affected by removing the DAM and only using one kind of activation function selected from ReLU, Sigmoid, and TanH. Results in Table II show significant decreases in detection accuracy, indicating the benefits of simultaneously using several activation functions. In addition, we performed experiments by computing the correlation coefficients between the averaged gradient values of each activation unit in the second and the fourth convolutional group for Sub-net #2 during the training stage. Table III shows the results, where we can observe that when two different activation functions are used, the correlation is lower. It indicates different activation units perform differently. Similar to ensemble learning systems where independence makes benefits, the diversity of the activation units improves the performance.

3) *Further investigation on parallel structure:* We would like to find out how the parallel structure affects the performance. To this end, we show experimental results with the following six cases. Case I: the use of only one sub-net by using Gabor, SRM linear, and SRM non-linear pre-processing filters all together ($N = 44$, where N is defined in Section III-C). Case II: the use of only one sub-net by using SRM linear and non-linear pre-processing filters together ($N = 28$). Case III: the use of Sub-net #1 and #2. Case IV: the use of Sub-net #1 and #3. Case V: the use of Sub-net #2 and #3. Case VI: the use of four sub-nets by replacing Sub-net #1 with two new sub-nets, where Gabor filters ($N = 8$) with scales $\sigma = 0.5$ and $\sigma = 1$ are respectively employed in the new sub-nets. The results for detecting S-UNIWARD are shown in Table IV. We can observe that, when only one sub-net is used, even though we increase the number of filters for pre-processing, it is not as effective as using more sub-nets with less number of filters. Besides, as the number of sub-nets increases, better performance can be obtained. To balance detection performance and model complexity, ReST-Net with three sub-nets is recommended.

V. CONCLUSION

Performing steganalysis using data-driven models have gradually evolved in the past few years. In this letter, we propose a new CNN architecture adapted from Xu-CNN with three parallel sub-nets in order to consider more pre-processed information. Our CNN also uses the diverse activation modules to activate the convolved data differently. The contributions of this paper rely on two aspects: (i) proposing diverse activation modules to learn steganographic artifacts in a diverse way; and (ii) showing that high-pass pre-processing operations in a parallel structure is better than simply stacking several filters together. As the proposed network is inspired by the *Inception module* [17], other effective structures for image classification, such as *ResNet* [18] and *DenseNet* [29], may help for the design of new steganalysis architectures.

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